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Clade-defining mutations in human H1N1 hemagglutinin protein from 2021-2023 have opposing effects on *in vitro* fitness and antigenic drift

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Running Head: Phenotypic effects of clade-defining H1N1 mutations

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Abstract

Seasonal influenza viruses frequently acquire mutations that have the potential to alter both virus replication and antigenic profile. Recent seasonal H1N1 viruses have acquired mutations to their hemagglutinin (HA) protein receptor binding site (RBS) and antigenic sites, and have branched into the clades 5a.2a and 5a.2a.1. Both clades demonstrated improved *in vitro* fitness compared with the parental 5a.2 clade as measured through plaque formation, infectious virus production in human nasal epithelial cells, and receptor binding diversity. Both clades also showed reduced neutralization by serum from healthcare workers vaccinated in the 2022-23 Northern Hemisphere influenza season compared to the vaccine strain. To investigate the phenotypic impact of individual clade-defining mutations, recombinant viruses containing single HA mutations were generated on a 5a.2 genetic background. The 5a.2a mutation Q189E improved plaque formation and virus replication, but was more efficiently neutralized by serum from individuals vaccinated in 2022-23. In contrast, the 5a.2a mutation E224A and both 5a.2a.1 mutations P137S and K142R impaired aspects of *in vitro* fitness but contributed significantly to antigenic drift. Surprisingly, the E224A mutation and not Q189E caused broader receptor binding diversity seen in clinical isolates of 5a.2a and 5a.2a.1, suggesting that receptor binding diversity alone may not be responsible for the phenotypic effects of the Q189E mutation. These data document an evolutionary trade-off between mutations that improve viral fitness and those that allow for the evasion of existing host immunity.

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41 **Importance**

42 Seasonal influenza is a significant contributor to global morbidity and mortality. To better
43 understand how these viruses evolve, recent seasonal H1N1 viruses were characterized and the
44 impact of individual mutational differences between the hemagglutinin proteins of these
45 viruses was assessed. Mutations that contribute to escape from preexisting immunity were
46 deleterious to virus replication in human nasal epithelial cell cultures, requiring the
47 accumulation of additional mutations to compensate for this loss of fitness. This adds to our
48 understanding of how influenza evolution balances replication fitness with antigenic drift.

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Introduction

Seasonal influenza viruses are among the largest causes of global respiratory infections. Interventions including antivirals and vaccinations have helped reduce the burden of disease, but influenza continues to infect roughly one billion people and cause the deaths of hundreds of thousands each year (1). One of the major challenges to combatting this disease is the genetic diversity present within these viruses, and the rapid pace of continued diversification. "Seasonal influenza" refers to two co-circulating genera of viruses from the *Orthomyxoviridae* family, influenza A virus (IAV) and influenza B virus, which can each be further subdivided into strains and subtypes. IAV is the predominant contributor to annual disease burden (2), and within IAVs both H1N1 and H3N2 subtypes co-circulate with relative prevalence changing from season to season (3).

Each IAV subtype consists of multiple clades as defined by the hemagglutinin (HA) sequence, with distinct genotypes and antigenic profiles (4). Mutations can alter numerous aspects of viral phenotype, including changes to both fitness and antigenic profile. Changes to fitness can include differences in receptor binding (5-8), virus replication (8-10), interactions with innate immunity (11), transmissibility (12-14), and other characteristics (15). Antigenic profile refers to the regions of a virus which are recognized by host antibodies, and changes to this profile through antigenic drift can alter how the virus appears to the immune system and result in a loss of antibody binding. Frequent mutations to viral antigenic sites contribute to our need for annual vaccination with updated strains whose antigenic profiles match those of currently circulating viruses (16). HA functions as the primary antigenic determinant of influenza and is also responsible for receptor binding and virus-host membrane fusion (17).

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These roles result in dual selective pressures on influenza to both evade existing host immunity and maintain or improve virological fitness by optimizing cell attachment and entry.

The H1N1 subtype has seen significant genetic diversification over the past several seasons, with multiple clades emerging and achieving varying degrees of prevalence. In the 2019-20 Northern Hemisphere (NH) influenza season the 5a.1 and 5a.2 clades both emerged from the 5a parental clade (18). The 5a.1 clade quickly predominated over 5a.2, and the 5a.2 clade was shown to have impaired *in vitro* fitness compared with 5a.1 which may have contributed to its limited spread (8).The 5a.2 clade rapidly acquired a series of HA mutations at the start of 2021 and branched into a novel subclade, 5a.2a, which quickly became the predominant H1N1 clade. 5a.2a is defined by mutations in the receptor binding site and antigenic site mutations A186T, Q189E, and E224A (H1 numbering). The A186T mutation has emerged previously in regionally circulating H1N1, including Ukraine (19), Tunisia (20), Turkey (21), Spain (22), Cuba (23), and the Czech Republic (24). Notably, Q189E had appeared as a clade-defining mutation in 5a.1 (8) but is new to the 5a.2 lineage. The E224A mutation has been shown to affect viral pathogenesis (25), cell tropism (26), and receptor binding (27, 28), and alanine already exists as the predominant residue at site 224 in swine H1 HA (25). Initial serological characterization indicates that 5a.2a is antigenically drifted from the 5a.2 clade (29, 30).

As the 5a.2a clade circulated in 2021 it continued to acquire mutations, and by 2022 the subclade 5a.2a.1 had emerged but remained a minority clade. It is defined by mutations including P137S and K142R, which occupy both the receptor binding site and antigenic site Ca2 . Both of these mutations have previously emerged in regional circulation (22, 31-33), and have

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been identified repeatedly in circulating swine H1 HA (34-37). Serological data suggests that this clade may be antigenically drifted from both the 2022-23 NH influenza vaccine strain and the 5a.2a clade (30).

To evaluate phenotypic changes to these recent H1N1 clades, experiments were performed to identify both antigenic and *in vitro* fitness differences between representative virus genotypes. Both 5a.2a and 5a.2a.1 clades were antigenically drifted from 5a.2 and demonstrated improved *in vitro* fitness. The mutations contributing to escape from vaccine induced immunity did so at the expense of infectious virus production and a mutation that improved virus replication increased virus recognition by serum from vaccinated individuals. These data highlight the evolutionary trade-off between acquiring mutations that allow for the evasion of existing immunity, while maintaining adequate virological fitness.

Results

Emergence of novel 5a.2 subclades

After the emergence of the 5a.1 and 5a.2 clades in 2019, 5a.1 quickly predominated and almost completely overtook 5a.2 by the end of 2020 (Figure 1A). Characterization of these viruses revealed a reduction in *in vitro* fitness for 5a.2 compared with 5a.1 (8), which may have contributed to the success of the latter. In 2021, a series of mutations emerged within the 5a.2 clade which were subsequently classified as a new clade, 5a.2a. 5a.2a spread rapidly and by 2022 overtook 5a.1 to become the primary H1N1 clade (Figure 1A). The mutations defining this new 5a.2a clade include multiple changes to antigenic sites, as well as several differences in the

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receptor binding site of the HA protein (Figure 1B). This suggests potential adaptations to the viral phenotype which may have improved fitness or antigenic profile, allowing it to circulate more widely than its 5a.2 parental clade.

The 5a.2a.1 clade, an offshoot of 5a.2a, also appeared around this time and slowly rose in prevalence as a sizable minority within H1N1 (Figure 1A). In addition to the mutations that define the 5a.2a clade, 5a.2a.1 contains several changes to both the receptor binding site and antigenic sites (Figure 1C), and initial serological data indicate that this clade may be antigenically drifted from 5a.2a (30), though its limited initial success suggests that any phenotypic changes acquired by this clade did not produce overwhelming advantages over 5a.2a.

To identify phenotypic differences between the 5a.2a and 5a.2a.1 clades, representative viruses were isolated from clinical samples of H1N1 collected from the Johns Hopkins Hospital during the 2022-23 Northern Hemisphere (NH) influenza season. These include two 5a.2a viruses, A/Baltimore/JH-22363/2022 and A/Baltimore/JH-22377/2022, and one 5a.2a.1 virus, A/Baltimore/JH-22400/2022. Two 5a.2a viruses were chosen to reflect intra-clade genetic diversity; A/Baltimore/JH-22377/2022 contains the T216A hemagglutinin mutation which appeared with regularity in this clade in 2022 and which has increased in prevalence in H1N1 since the 2022-23 season. In addition, a 5a.2 virus from the 2019-20 NH season, A/Baltimore/R0675/2019, was included as a comparison group for the parental 5a.2 clade.

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Recent H1N1 clades show improved in vitro *fitness compared with the parental 5a.2 clade*

To compare *in vitro* fitness between these clades, plaque formation and viral replication were assessed. Both 5a.2a and 5a.2a.1 viruses produced significantly larger plaques on MDCK cells compared with the 5a.2 parental clade (Figure 2A, 2B). Additionally, the 5a.2a virus A/Baltimore/JH-22363/2022 produced significantly larger plaques than the 5a.2a.1 virus A/Baltimore/JH-22400/2022, though neither were significantly different from the 5a.2a virus A/Baltimore/JH-22377/2022 which contains the T216A mutation (Figure 2B). Low-MOI growth curves were then performed on human nasal epithelial cell (hNEC) cultures, which are primary differentiated cell cultures grown at an air-liquid interface, and are an *in vitro* model of the upper respiratory tract (38-42). Both 5a.2a viruses, but not 5a.2a.1, replicated to significantly higher titers at multiple timepoints compared with 5a.2 (Figure 2C). To compare total infectious virus production, the area under the curve was calculated for each growth curve, which confirmed that 5a.2a, but not 5a.2a.1, produced significantly more infectious virus compared with 5a.2 (Figure 2D). Differences in replication were observed primarily at timepoints after peak virus titer had been reached, so to investigate this phenotype, virus production was separated into pre-peak and post-peak sections. Total virus production pre-peak was not significantly different between any of the clades (Figure 2E), however post-peak virus production was significantly higher for both recent H1N1 clades compared with the 5a.2 clade (Figure 2F).

Recent H1N1 clades are antigenically drifted from their ancestral clade

Antigenic changes were assessed through virus neutralization assays using serum from the Johns Hopkins Centers for Excellence in Influenza Research and Response (JH-CEIRR) 2022

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158 cohort. This cohort includes serum samples from healthcare workers both pre- and post-
159 vaccination with the annual influenza vaccine, and participants include both men and women
160 with ages ranging from 22-75. At both pre- and post-vaccination timepoints, the 5a.2a and
161 5a.2a.1 clades were less well neutralized by serum antibodies compared with the vaccine strain
162 for that season, a virus from the 5a.2 clade (Figure 3A, 3B). The 5a.2a.1 clade also appeared to
163 be antigenically drifted from 5a.2a, showing a significant reduction in neutralization both pre-
164 and post-vaccination compared with at least one of the representative 5a.2a viruses.
165 Interestingly, vaccination produced a comparable boost in neutralizing antibody titers across
166 the virus panel; the fold change in neutralizing titers was similar for all four viruses tested
167 (Figure 3C), and seroconversion rates were not significantly different (Figure 3D).

168

169 *Identifying the phenotypic contributions of individual clade-defining mutations*

170 While the initial characterization of representative clinical isolates from these clades
171 suggests important phenotypic changes associated with recent clade-defining mutations, the
172 impact of individual gene segments and mutations cannot be determined by characterizing
173 clinical virus isolates alone. To elucidate the contribution of individual genes and mutations, an
174 infectious clone of the A/Baltimore/R0675/2019 5a.2 virus was generated as previously
175 described (43) by creating eight plasmids containing individual gene segments of the virus (5,
176 44, 45). To confirm that the virus cloning and rescue process alone did not alter viral
177 phenotype, a rescued A/Baltimore/R0675/2019 virus was compared with the genetically
178 identical wildtype virus and both demonstrated similar *in vitro* fitness as judged by plaque sizes
179 (Figure 4A and 4B) and replication on hNEC cultures (Figure 4C to 4F).

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To confirm that changes to the hemagglutinin protein were responsible for the observed differences between 5a.2 and 5a.2a, a reassortant virus was generated containing seven gene segments from the 5a.2 virus and an HA segment from 5a.2a (A/Sydney/5/2021). This reassorted virus generated significantly larger plaques on MDCK cells (Figure 4A and 4B) and replicated to significantly higher titers on hNEC cultures compared with the 5a.2 virus (Figure 4C and 4D). Virus replication reflected the trend seen in clinical isolates, with differences in total virus production being driven by increases in post-peak virus production (Figure 4E and 4F).

After confirming that the HA protein was driving differences between clades, the contributions of individual clade-defining mutations were assessed. Viruses were generated using the seven segments from the 5a.2 virus along with 5a.2 HA segments containing individual point mutations from either the 5a.2a (A186T, Q189E, and E224A) or 5a.2a.1 (P137S and K142R) clades. The Q189E mutation from the 5a.2a clade significantly increased plaque size on MDCK cells compared with the 5a.2 virus, while the E224A mutation from 5a.2a and the P137S mutation from 5a.2a.1 both reduced plaque size (Figure 5A, 5B).

Growth curves on human nasal epithelial cells showed that every point mutant resulted in significantly different viral titers at multiple timepoints post-infection compared with the 5a.2 virus (Figure 5C). Area under the curve analysis revealed that total virus production was reduced for all viruses except for the Q189E mutant (Figure 5D), with every single mutant virus producing significantly less virus than the 5a.2 virus before peak titer (Figure 5E). Post-peak production, however, was increased for Q189E compared with 5a.2 (Figure 5F), mirroring the trend seen in both the clinical isolates and the 5a.2a reassorted virus. Both E224A and the two

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5a.2a.1 mutations (P137S and K142R) produced significantly less virus post-peak compared with 5a.2 (Figure 5F).

Differences in receptor binding between recent H1N1 clades

The changes observed to *in vitro* fitness of recent clades, combined with the location of clade-defining mutations on the receptor binding site, suggested a role for receptor binding as a mediator for fitness changes. To investigate receptor binding diversity, Consortium for Functional Glycomics glycan microarrays were performed on both the clinical isolate panel and the mutant virus panel. These glycan arrays contain over 100 terminally sialylated glycans including ones with either alpha 2,3 or alpha 2,6 linkages. Receptor binding profiles for both clinical viruses and point mutants demonstrate a preference for alpha 2,6 terminally sialylated glycans, the canonical human influenza receptor, and show differences in both quantity and identity of bound glycans (Figure 6A, 6B). To visualize these differences, a heat map was generated of all alpha 2,6 glycans included in the array, showing relative binding intensity for each virus. While there was some overlap in bound glycans among all viruses (indicated by a blue circle), both 5a.2a and 5a.2a.1 viruses bound to additional glycans not bound by the parental 5a.2 clade (indicated by a green circle), and these glycans were also bound by the E224A mutant virus (Figure 6C, CD). This suggests that the E224A mutation is responsible for this increase in receptor binding diversity of recent H1N1 clades. Additionally, the K142R mutation from the 5a.2a.1 clade was shown to abrogate binding to a specific glycan (ID number 560), which was strongly bound by all other viruses (indicated by a purple circle). The Q189E mutation that increased *in vitro* replication fitness did not show major changes in glycan

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binding pattern, suggesting that altered receptor binding diversity is most likely not the reason for the improved fitness of viruses containing this mutation. These results show that recent H1N1 mutations have produced altered receptor binding diversity but that receptor binding diversity alone does not correlate with overall viral fitness.

Individual HA mutations can reduce or increase neutralization by serum from vaccinated individuals

To assess the contributions of individual clade-defining mutations on antigenic drift, neutralization assays were performed on the point mutant virus panel using post-vaccination serum from the 2022 JH-CEIRR healthcare worker cohort. The E224A, P137S, and K142R mutations all significantly reduced serum neutralization compared with the vaccine strain, while the addition of the Q189E mutation actually increased virus neutralization by serum antibodies (Figure 7). These data indicate that the E224A and 5a.2a.1 mutations, while potentially deleterious to the virus from an *in vitro* fitness perspective, may have been selected due to their contribution to antigenic drift and evasion of existing immunity. Conversely, the Q189E mutation increases neutralization by serum compared to the vaccine strain whereas the same mutation improved virus fitness *in vitro*. Taken together, these data indicate that the presence of mutations at multiple antigenic sites allows for the tolerance of a mutation, Q189E, that increases serum neutralization, particularly when that mutation can improve *in vitro* virus fitness.

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Discussion

The emergence of the 5a.2a clade of H1N1, and the speed with which it predominated, was surprising considering the failure of its parental clade, 5a.2 (Figure 1A). The 5a.2 clade had significantly reduced *in vitro* fitness compared with the cocirculating 5a.1 clade and their parental 5a clade, including reduced receptor binding diversity, replication, and plaque formation (8). The contrasting success of its progeny 5a.2a clade suggested that there were phenotypic advantages conferred by the 5a.2a clade-defining mutations. Indeed, *in vitro* fitness assays reveal that 5a.2a forms significantly larger plaques (Figure 2A, 2B), replicates significantly better in human nasal epithelial cell (hNEC) cultures (Figure 2C-2F), and has increased receptor binding diversity (Figure 6C) compared with 5a.2. The 5a.2a clade was also antigenically drifted from 5a.2, as demonstrated by neutralization assays using serum from healthcare workers (Figure 3A, 3B). These advantages likely contributed to the ability of 5a.2a to overtake the other co-circulating H1N1 clades.

Three clade-defining mutations for 5a.2a, A186T, Q189E, and E224A, occupy both the receptor binding site (RBS) and antigenic site Sb (Figure 1B), suggesting a key role in fitness differences. When characterized, it was shown that only Q189E was able to independently increase *in vitro* fitness, while E224A actually reduced plaque size and total virus production in hNEC cultures (Figure 5A-F). Interestingly, E224A was the only 5a.2a mutation that contributed to antigenic drift, while Q189E actually resulted in increased virus neutralization (Figure 7). Q189E also appeared as a clade-defining mutation of the 5a.1 clade that predominated from 2020-2021 (Figure 1A) and was present in the 2021-22 Northern Hemisphere (NH) influenza vaccine (46). Exposure to 5a.1 through either natural infection or vaccination may have

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generated antibody responses against this epitope, and the healthcare workers whose serum was tested above (Figures 3 and 7) would have received the 2021-22 NH vaccine due to workplace vaccination requirements. The 224 residue, on the other hand, has been glutamic acid since the emergence of 2009pdm H1N1 (47), and its switch to alanine in 2021 would have been the first population-level exposure to a new amino acid at this site. The phenotypic contributions of Q189E and E224A highlight the evolutionary balance between mutations that alter fitness and those that contribute to antigenic drift. Given the overlap between the RBS and key antigenic sites, a benefit to one component may come at a cost to the other. The accrual of multiple mutations with contrasting advantages and disadvantages may allow the virus to improve its phenotype while mitigating the deleterious effects of any one change.

Receptor binding specificity is a major factor by which hemagglutinin contributes to virus replication and it has been shown to correlate with viral fitness (5-8). The reduction of *in vitro* fitness of the 5a.2 clade was associated with a loss of receptor binding diversity (8), suggesting that differences in cell attachment may have contributed to the poor circulation of 5a.2. It was therefore hypothesized that the improved *in vitro* fitness of the 5a.2a clade would be associated with an increase in receptor binding diversity, and this was confirmed by glycan microarray experiments (Figure 6). Since the Q189E mutation was shown to be the driver of the improved *in vitro* fitness phenotype, this was the mutation that was expected to contribute to the increased receptor binding diversity. Surprisingly, the effect of Q189E on receptor binding was negligible, while the E224A mutation appears to be the primary cause of the increased diversity (Figure 6). Since the stark effect of Q189E on viral replication and plaque formation was not correlated with an increase in receptor binding diversity, its mechanism for producing

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the observed *in vitro* fitness changes is not yet completely understood. The glycan microarray used to measure binding diversity consists of synthetic glycans; the mutations tested above may produce differences in receptor binding that are not able to be captured using a synthetic microarray. An array using naturally-derived glycans may reveal differences more specific to the human respiratory tract (48, 49). Glycan microarrays may also overlook differences in low-affinity and non-canonical receptor interactions which have been shown to contribute to viral entry (50, 51). It is also possible that receptor affinity might be altered by one or more of these mutations without producing a significant broadening of receptor binding diversity. Changes in affinity may also explain the differences in replication seen pre-peak versus post-peak for these viruses.

The 224 residue, and specifically the E224A mutation, has previously been studied for its effect on RBS confirmation. It was noted shortly after its emergence that the 2009 pandemic H1N1 may have reduced stability in its interactions between the 190-helix and 220-loop compared with the 1918 H1N1 due to the amino acids at positions 216 and 224 (52). Shang *et al.* modeled differences between these viruses and found that the E224A mutation, in combination with S183P, stabilized interactions between these two regions (27). The addition of E224A to the pdm2009 genetic background was shown to increase binding affinity to fetuin (53) and alter cell tropism (26), though it replicated to lower titers on human airway epithelial cells and had no impact on transmission in ferrets (41). Jayaraman *et al* predicted in 2011 that changing key RBS residues, including E224, may improve receptor binding affinity by creating a more hydrophobic interaction network in this region of the RBS (25, 53, 54). They generated a pdm2009 virus with S183P, A186T, and E224A and demonstrated that it had improved receptor

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affinity compared with the original pdm2009. Interestingly, all three of these mutations have since become the predominant residues at their respective sites, indicating that this constellation of amino acids is advantageous to the virus. E224A only emerged once these other mutations were also present in circulating H1, and it is possible that there are fitness advantages of E224A that only manifest in the context of these other mutations.

Similarly, while A186T did not appear to produce any apparent *in vitro* fitness advantages on its own, it may produce a synergistic effect when combined with E224A. It has been shown that some mutations only contribute to phenotypic changes when combined with a second mutation, and that the receptor binding phenotype of E224A can be enhanced by additional mutations (25, 53). Experiments that test combinations of recent clade-defining mutations are needed to elucidate any synergistic effects of these amino acid changes.

In addition to assessing differences between the 5a.2a and 5a.2 clades, this study also investigated characteristics of the 5a.2a.1 clade that emerged from 5a.2a. Unlike 5a.2a, 5a.2a.1 did not rapidly predominate within circulating H1N1, but rather gradually increased in prevalence (Figure 1A). The 5a.2a.1 clade was shown to replicate at least as well as 5a.2, and it produced significantly larger plaques and had increased receptor binding diversity (Figures 2 and 6). Both of its major clade-defining mutations, however, resulted in a reduction in performance on one or more *in vitro* fitness assays (Figures 5 and 6). The benefits of these mutations appear to be largely antigenic, as they both significantly reduced serum neutralization when added to the 5a.2 genetic background (Figure 7), and likely contributed to the overall antigenic drift seen for 5a.2a.1 (Figure 3A, 3B).

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While the phenotypes of individual 5a.2a.1 mutations were characterized here by adding them to a 5a.2 genetic background, circulating 5a.2a.1 viruses also contain all of the clade-defining mutations associated with 5a.2a. This likely explains the difference in *in vitro* fitness observed between the clinical 5a.2a.1 virus and its individual clade-defining mutations. The deleterious effects of the 5a.2a.1 mutations are likely ameliorated by advantages conferred by one or more 5a.2a mutations, allowing this virus to have adequate fitness while preserving the antigenic changes resulting from the new mutations. This exemplifies the evolutionary balancing act between mutations conferring fitness benefits and those that allow the virus to evade adaptive immunity. One model of influenza evolution proposes the cyclical emergence of mutations with varying levels of receptor avidity influenced by pressure to escape immunity (55), and earlier surveillance of circulating H1N1 since 2009 has shown that mutations increasing receptor avidity were followed by those that reduced avidity (28). An analogous process may be driving recent H1N1 evolution, with the Q189E mutation of 5a.2a increasing viral fitness sufficiently to allow for subsequent 5a.2a.1 mutations driving antigenic drift.

This process can be understood through epistasis, or the impact of secondary mutations on an individual mutant phenotype. Epistatic processes are thought to be a major component of influenza evolution, due in part to overlapping functional regions of hemagglutinin and the frequency of pleiotropic mutations (56, 57). Epistasis in the RBS specifically has been shown to allow for significant sequence diversity and antigenic escape while maintaining viral fitness (58). The striking difference between individual and combined mutations shown here, as well as the recent temporal clustering of RBS mutations, points to a significant epistatic component to contemporary H1N1 clade-defining mutations. The synergistic effect of these mutations,

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354 including their contribution to receptor avidity and affinity, should be investigated to determine
355 how the E224A mutation is impacted by the 190-helix mutations, and whether positive epistasis
356 is observed between Q189E and the 5a.2a.1 mutations.

357

358 **Methods**

359 **Ethics statement and human subjects**

360 Serum used in this study was obtained from healthcare workers recruited during the annual
361 Johns Hopkins Hospital employee influenza vaccination campaign in the Fall of 2022 by the
362 Johns Hopkins Centers for Influenza Research and Response (JH-CEIRR). Pre- and post-
363 vaccination (~28 day) human serum were collected from subjects, who provided written
364 informed consent prior to participation. The JHU School of Medicine Institutional Review Board
365 approved this study, IRB00288258. Virus was isolated for this study from deidentified influenza
366 A virus H1N1 positive samples, collected from patients who provided written informed consent
367 during the 2022-23 influenza season at the Johns Hopkins Hospital, under the JHU School of
368 Medicine Institutional Review Board approved protocol, IRB00091667.

369 **Cell cultures**

370 Madin-Darby canine kidney (MDCK) cells (provided by Dr. Robert A. Lamb) and MDCK-SIAT cells
371 (provided by Dr. Scott Hensley) were maintained in complete medium (CM) consisting of
372 Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum, 100
373 units/ml penicillin/streptomycin (Life Technologies) and 2 mM Glutamax (Gibco) at 37 °C and

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374 5% CO_2. Human nasal epithelial cells (hNEC) (PromoCell) were cultivated as previously
375 described (5, 8).

376 **Viral RNA Extraction and Multi-segment Amplification by Reverse-Transcriptase Polymerase**
377 **Chain Reaction**

378 Viral nucleic acid was extracted from samples as previously described (59) using the Chemagic
379 Viral RNA/DNA Kit following the manufacturer's instructions (Revvity). Samples were eluted
380 into 60µL of elution buffer and used directly after extraction in PCR. The whole genome was
381 amplified using a single multi-segment 1-step reverse-transcription polymerase chain reaction
382 (RT-PCR) with the MBTuni-12 [5'-ACGCGTGATCAGCAAAAGCAGG] and MBTuni-13 [5'-
383 ACGCGTGATCAGTAGAAACAAGG] primer sets for Influenza A (59, 60) with the SuperScript III
384 One-Step RT-PCR kit as per manufacturers guidelines (Invitrogen).

385 **Library Preparation and Next-Generation Sequencing and Virus Genome Assembly**

386 After amplification, concentrations were checked using the Qubit 2.0 fluorometer
387 (ThermoFisher Scientific). 2µL of PCR product was prepared for sequencing using the native
388 barcoding genomic DNA kit (EXP-NBD196) and the NEBNext ARTIC Library Prep kit, following
389 the manufacturer's instructions (New England BioLabs / Oxford Nanopore Technologies). 50ng
390 of the barcoded library was sequenced using the R9.4.1 flow cell on a GridION (Oxford
391 Nanopore Technologies). Resulting fastq files were demultiplexed using the artic_guppyplex
392 tool (Artic version 1.2.2). Nucleotide sequence assembly was performed using the FLU module

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of the Iterative Refinement Meta-Assembler (IRMA version 1.0.2), using IRMA's default settings, which include a minimum average quality score of 24 and a site depth of 100 (59, 61).

Clinical virus sequences are accessible through GISAID with the following IDs: EPI_ISL_17617226 for A/Baltimore/R0675/2019, EPI_ISL_17626342 for A/Baltimore/JH-22363/2022, EPI_ISL_17626298 for A/Baltimore/JH-22377/2022, and EPI_ISL_17626340 for A/Baltimore/JH-22400/2022.

Viral isolation

Viral stock preparation was performed as described previously (8, 62) using nasopharyngeal swabs from influenza H1N1-positive individuals during the 2022–2023 Northern Hemisphere influenza season at JHH.

Virus Cloning and Plasmids

Virus cloning was performed as previously described (5, 10, 63). Viral RNA was extracted using the QIAamp viral RNA mini extraction kit and converted to cDNA using universal influenza A primers. Individual segments were amplified from cDNA with Phusion High-Fidelity DNA Polymerase (New England BioLabs) using segment-specific primers, which also added restriction enzyme sites to the ends of the segments. For all segments, the pHH21 plasmid was used (64, 65). Both the plasmid and individual segments were digested separately with restriction enzymes (either BsmBI or BsaI, New England BioLabs) and then combined and ligated together with T4 DNA ligase (New England BioLabs). Ligated plasmids were transformed into DH5α

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bacteria (Thermo Fisher) and grown overnight on LB plates containing carbenicillin to select for colonies that successfully received the plasmid. Individual colonies were isolated and grown, and plasmid was extracted using a Qiagen QIAprep Spin Miniprep Kit. Plasmid sequences were verified using restriction enzyme digestion and whole plasmid sequencing, which was performed by Plasmidsaurus using Oxford Nanopore Technology with custom analysis and annotation. Bacteria expressing the correct plasmid were grown in bulk and high-purity plasmid was extracted using ZymoPURE Plasmid Midiprep kit (Zymo research) and verified by sequencing of the coding region of the influenza virus segment.

For generating plasmids containing specific point mutations, mutagenic primers were designed using the New England Biolabs online software NEBaseChanger (primer sequences are available at doi: XXXX. The Q5 Site-Directed Mutagenesis Kit (New England BioLabs) was then used with the A/Baltimore/R0675/2019 hemagglutinin plasmid to produce plasmids with the desired mutations. Plasmids were transformed, extracted, and confirmed as described above.

Recombinant Virus Production

Recombinant viruses were generated using the 12 plasmid reverse genetics system described previously (5, 44, 45).

Tissue culture infectious dose 50 (TCID50)

TCID50 assays were performed on MDCK-SIAT cells as previously described (8, 10, 66). The 50% tissue culture infectious dose was calculated as previously described (64, 67).

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Plaque assay

Plaque assays were performed and analyzed as previously described (8, 10) using MDCK cells grown in complete medium to confluence in 6-well plates. Overlay media was phenol-red free DMEM supplemented with .3% BSA (Sigma), 100U/ml pen/strep (Life Technologies), 2mM Glutamax (Gibco), 5mM HEPES buffer (Gibco) 5µg/ml N-acetyl trypsin (Sigma) and 1% cellulose.

Viral growth curves

hNEC growth curves were performed at an MOI of 0.01 TCID50 units per cell on fully differentiated hNECs grown at an air–liquid interface as described previously (8, 10, 66). Infectious virus concentration in collected media was determined through TCID50.

Partial virus purification and labeling with Alexa Fluor 488

For each virus analyzed using glycan arrays, 100 mL of virus stock was grown using the method referenced above, and viruses were partially purified through ultracentrifugation and labeled with Alexa Fluor 488 as previously described (8). Labeled virus was stored at −80 °C, and TCID50 was performed to confirm virus activity and concentration. Titer of labeled viruses were: 4.20E+8 TCID50/mL (A/Baltimore/R0675/2019), 1.33E+8 TCID50/mL (A/Baltimore/JH-22363/2022), 1.70E+8 TCID50/mL (A/Baltimore/JH-22377/2022), 5.37E+8 TCID50/mL (A/Baltimore/JH-22400), 1.58E+8 TCID50/mL (A/Baltimore/R0675/2019 with A186T), 1.95E+8 TCID50/mL (A/Baltimore/R0675/2019 with Q189E), 4.20E+7 TCID50/mL (A/Baltimore/R0675/2019 with E224A), 6.17E+7 TCID50/mL (A/Baltimore/R0675/2019 with P137S), and 2.86E+7 TCID50/mL (A/Baltimore/R0675/2019 with K142R).

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Glycan array

The Consortium for Functional Glycomics version 5.5 microarray, containing 562 glycans in replicates of 6, was used for glycan arrays as previously described (8, 68). Microarray slides were submerged in 100 ml of PBS wash buffer (phosphate-buffered saline containing 0.005% Tween-20) for five minutes to re-hydrate. 70 µl of undiluted sample were loaded onto the slide. A cover slip was placed over the slide and incubated at room temperature for one hour in darkness, after which the cover slip was removed and the slide was washed four times in 100 ml of PBS wash buffer, then PBS, then deionized water. Slides were then dried before being read in a GenePix microarray scanner.

For each set of six replicates, only the median four values were used in analysis. A graph of the raw data was visually analyzed to confirm selective binding of terminally sialylated glycans, after which data was filtered to only include these glycans for a total of 139 glycans. Data was then normalized to a maximum value of 100, and organized by structure using the Glycan Array Dashboard (GLAD) (69). Both the heat map and structures of individual glycans were generated through GLAD.

Serum neutralization assay

Pre- and post-vaccination human serum was treated with Receptor Destroying Enzyme II (RDE, Hardy Diagnostics) and neutralization assays were performed on MDCK-SIAT cells as previously described (8, 70), performed in quadruplicate and using 100 TCID50 of virus per well.

Statistical analyses

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All statistical analyses were performed in GraphPad Prism 9.1.0. Growth curves were analyzed using 2-way ANOVA with Tukey post-hoc test. Area under the curve (AUC) was calculated for each growth curve replicate and AUC values were compared using one-way ANOVA for differences in total virus production. Plaque assays were analyzed using Kruskal–Wallis ANOVA with Dunn's multiple comparison test. Neutralization assays and fold-change in neutralizing titers were analyzed using repeated-measures one-way ANOVA with Tukey post-hoc, and differences in seroconversion between clades were analyzed using a Chi-square test.

Data availability

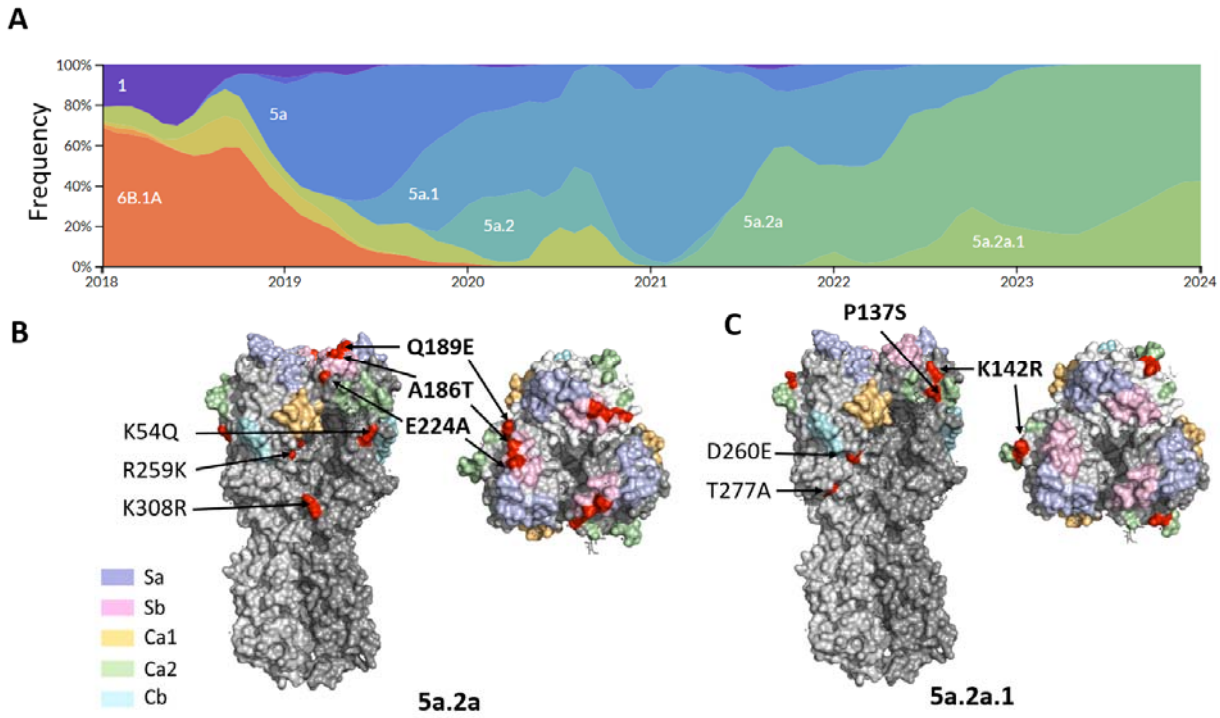
The datasets used and/or analyzed in the current study are available through the Johns Hopkins Research Data Repository at doi: XXXX

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Acknowledgements

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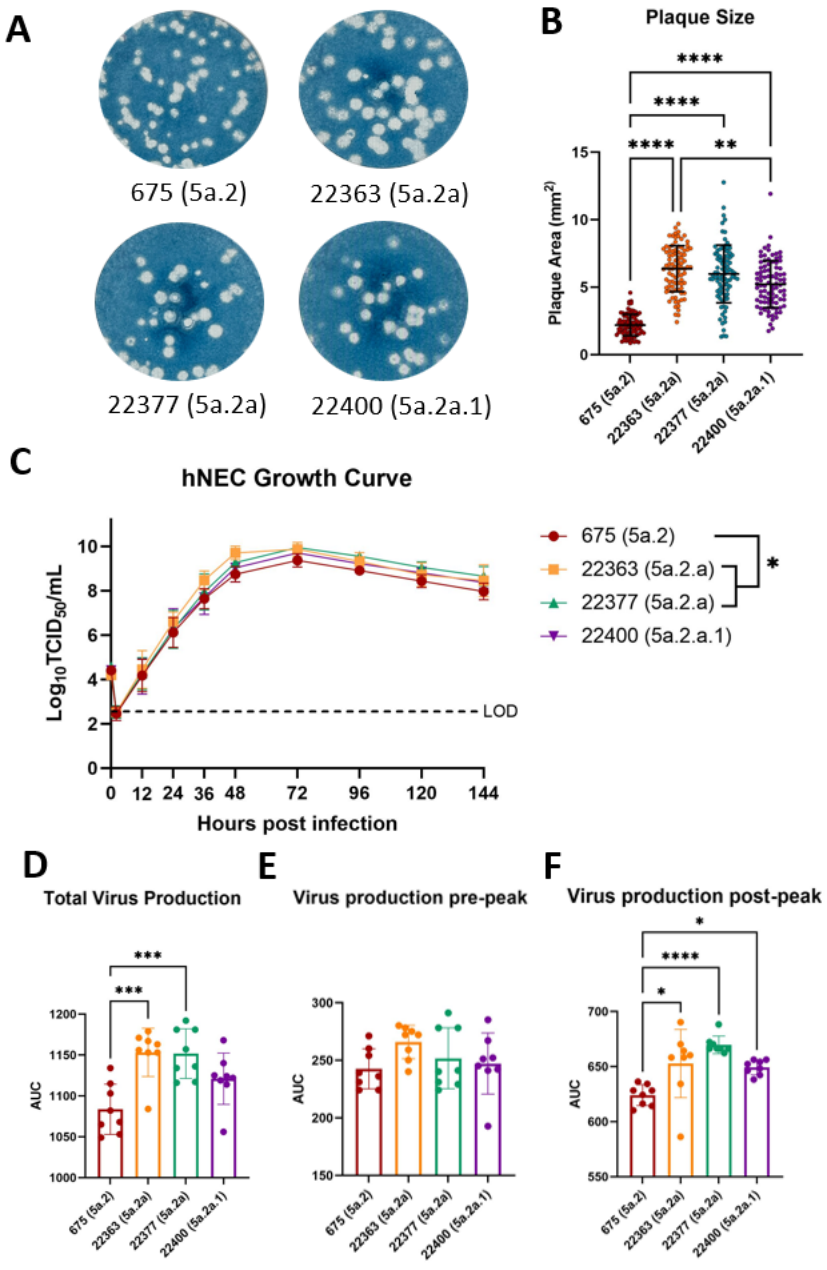


497

498 Figure 1: Emergence of seasonal H1N1 clades. A) Frequencies of circulating clades of H1N1 since
499 2018, generated using the NextStrain pipeline (47). Clade 5a.2a emerged in 2021 and quickly
500 predominated, followed by 5a.2a.1 which slowly gained prevalence. Clade-defining mutations of
501 B) 5a.2a and C) 5a.2a.1 mapped on the hemagglutinin trimer using PyMOL (PDB: 3LZG), with
502 antigenic sites shaded. In addition to the mutations shown, 5a.2a.1 also contains all the clade-
503 defining mutations of 5a.2a. Mutations surrounding the receptor binding site are in bold.

504

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505

506 Figure 2: Characterization of *in vitro* replication fitness of circulating H1N1 viruses. A) Plaque
507 formation on MDCK cells of representative viruses from circulating clades compared with their
508 parental clade (5a.2). B) Recently circulating clades formed significantly larger plaques than
509 their parental clade (Kruskal–Wallis ANOVA with Dunn's multiple comparisons test). Growth
510 curves on human nasal epithelial cells reveal an increase in C) the kinetics of infectious virus

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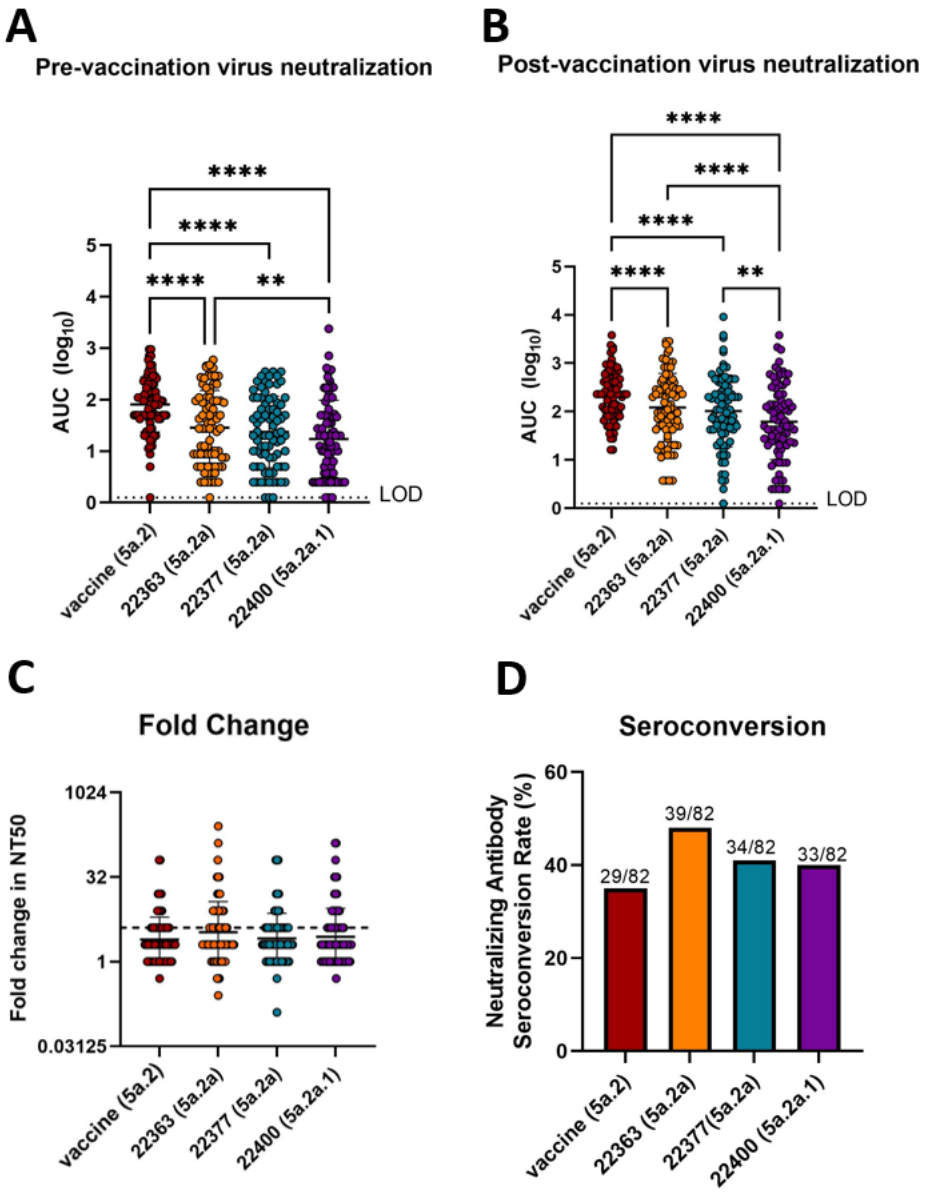
511 production and D) total virus production for the 5a.2a clade compared with 5a.2. Infectious

512 virus production differences are not present in E) pre-peak titers (peak=72 hpi) but are driven by

513 F) post-peak viral titers. One-way (AUC) or two-way (growth curve) ANOVA with Tukey post-hoc.

514 * p<0.05, ** p<0.01, *** p<0.001, **** p<0.0001

515



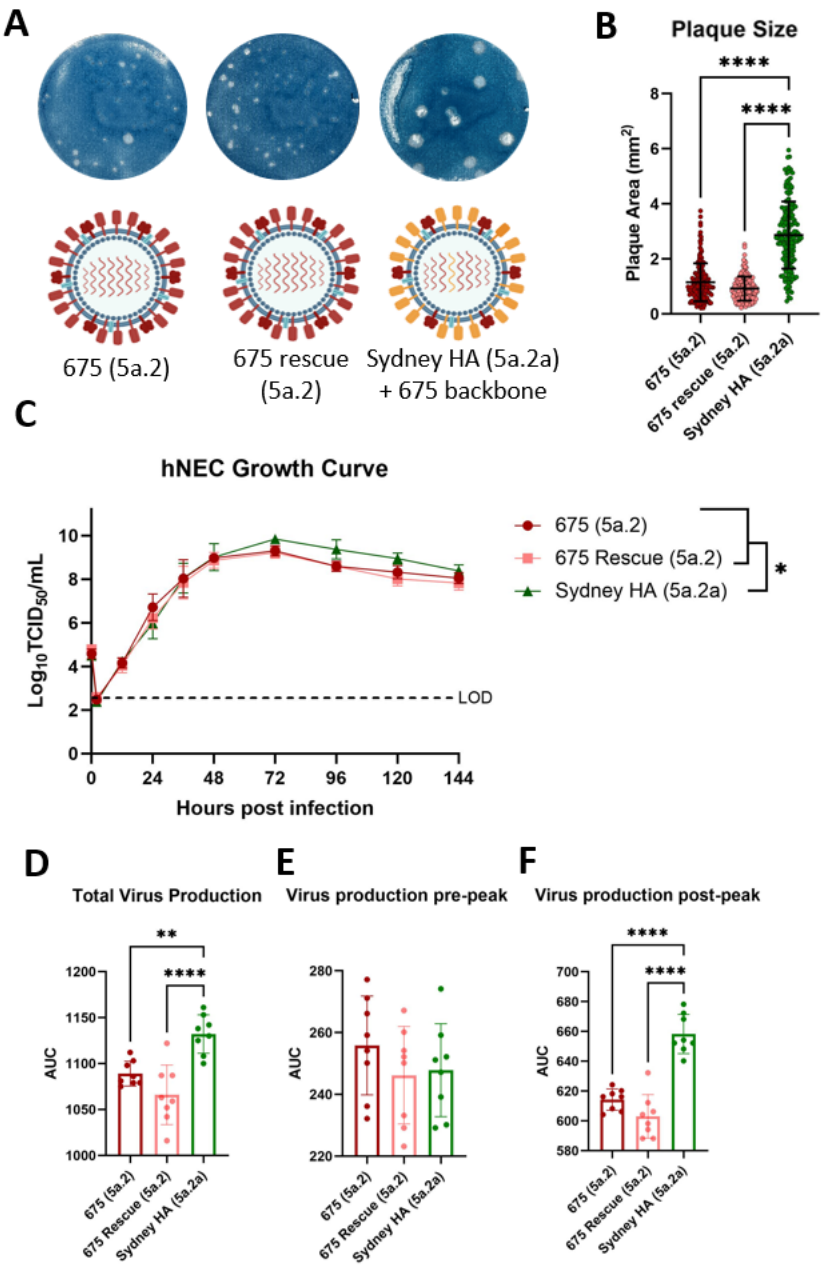
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517 Figure 3: Neutralization of vaccine and circulating viruses by human serum. Virus neutralization
518 was assessed using serum from a cohort of healthcare workers both A) pre- and B) post-
519 vaccination with the 2022-23 Northern Hemisphere influenza vaccine. The virus panel included
520 representative viruses from 5a.2a and 5a.2a.1 clades and the H1N1 vaccine strain, a 5a.2-clade
521 virus (repeated measures one-way ANOVA with Tukey post-hoc). 5a.2a viruses were significantly
522 less well neutralized than the vaccine virus, and 5a.2a.1 had a further reduction from 5a.2a in
523 both serum panels. C) Fold change in neutralizing antibody titers after vaccination was similar
524 for all viruses (repeated measures one-way ANOVA with Tukey post-hoc). D) Percent of serum
525 donors who seroconverted (showed at least a fourfold increase in neutralizing titers post-
526 vaccination) was similar for each virus (Chi-square test). N=82. ** $p<0.01$, **** $p<0.0001$

527

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528

529 Figure 4: The contribution of the 5a.2a HA segment on *in vitro* replication fitness. A) A 1:7

530 reassortant virus was generated containing the HA from a 5a.2a virus (A/Sydney/5/2021) with

531 all seven remaining segments from a 5a.2 virus (A/Baltimore/R0675/2019), and plaque assays

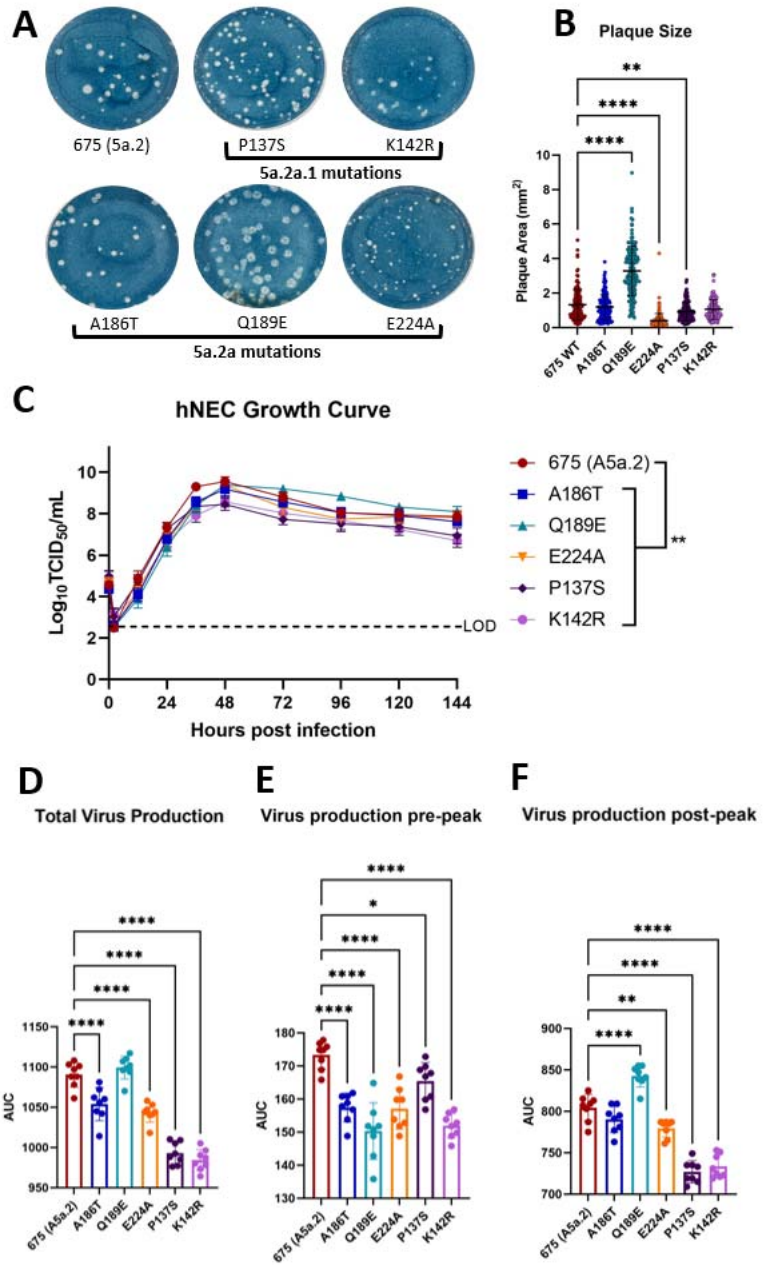
532 were performed on MDCK cells. B) The reassortant virus produced significantly larger plaques

533 than the wild type 5a.2 virus and a control rescue 5a.2 virus (Kruskal–Wallis ANOVA with Dunn's

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multiple comparisons test). The reassortant virus replicated to a greater extent in hNEC cultures C), producing a greater amount of D) total infectious virus particles compared to the parental virus. Infectious virus production E) pre-peak was not different (peak=72 hpi) but more infectious virus was produced F) post-peak with the 5a.2a HA segment containing virus. One-way (AUC) or two-way (growth curve) ANOVA with Tukey post-hoc. ** $p<0.01$, **** $p<0.0001$

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541

542 Figure 5: Determining the impact of individual clade-defining mutations in HA on *in*

543 *vitro* replication fitness. Viruses were generated using a 5a.2 background

544 (A/Baltimore/R0675/2019) containing individual point mutations from recent H1N1 clades

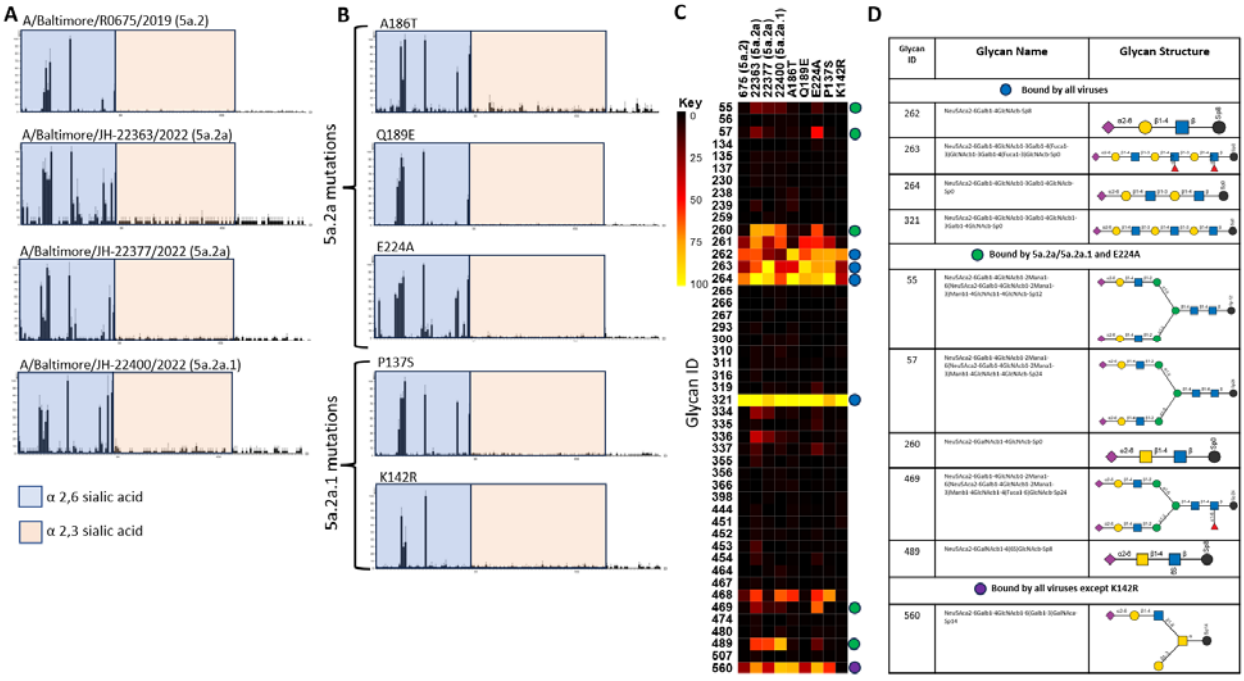
545 (5a.2a and 5a.2a.1). A) Representative images of MDCK plaque assays. B) The Q189E mutation

546 from 5a.2a significantly increased plaque area, while E224A and P137S reduced plaque area

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547 (Kruskal–Wallis ANOVA with Dunn's multiple comparisons test). C) Viral growth curves on
548 human nasal epithelial cells reveal significant temporal differences in replication between the
549 5a.2 virus and all point mutant viruses (Two-way ANOVA with Tukey post-hoc). D) Total virus
550 production is reduced with all mutations except for Q189E. E) Pre-peak infectious virus titer is
551 reduced for all point mutant viruses (peak=48 hpi), however virus production F) post-peak is
552 only reduced for E224A, P137S, and K142R, while the Q189E mutation increases total virus
553 production (One-way ANOVA with Tukey post-hoc). * p<0.05, ** p<0.01, *** p<0.001, ****
554 p<0.0001

555



556

557 Figure 6: Comparison of receptor binding activity via the Consortium for
558 Functional Glycomics (CFG version 5.5) glycan microarray. Receptor binding profiles of H1N1
559 clinical isolates A) and point mutants on a 5a.2-clade background B) show preference for alpha

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2,6 terminally sialylated glycans, and differences in individual glycans bound. X axis depicts individual glycans included in the array, and Y axis depicts the relative fluorescence units (RFUs), a metric of virus binding, normalized to 100. C) heat map of normalized RFU values for each glycan, for each virus. A yellow square indicates maximum binding by that virus. Glycans of note are marked with colored circles; blue are glycans that were well-bound by all viruses, green are glycans bound well by viruses from the 5a.2a and 5a.2a.1 clades as well as the E224A point mutant virus, and purple is a glycan that was well-bound by all viruses except the K142R point mutant. D) Structures and names of glycans of note.

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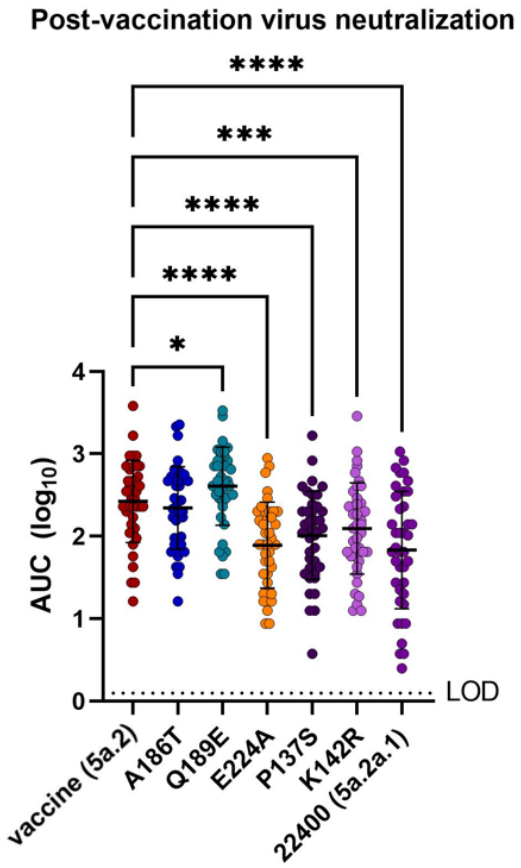


Figure 7: The impact of individual mutations on antigenic drift was assessed on a subset of post-vaccination serum from the 2022-23 Northern Hemisphere influenza vaccine healthcare worker cohort. The E224A, P137S, and K142R mutations reduced virus neutralization by serum antibodies, while the Q189E mutation increased neutralization and A186T had no discernable effect (repeated measures one-way ANOVA with Dunnett post-hoc using A/Victoria as comparison group). N=40. * $p<0.05$, *** $p<0.001$, **** $p<0.0001$

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